ALBERTSCHOOL

MINES PARIS | PSL

CORPORATE STRATEGY

Session 8: Data Strategy

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Data Strategy Introduction

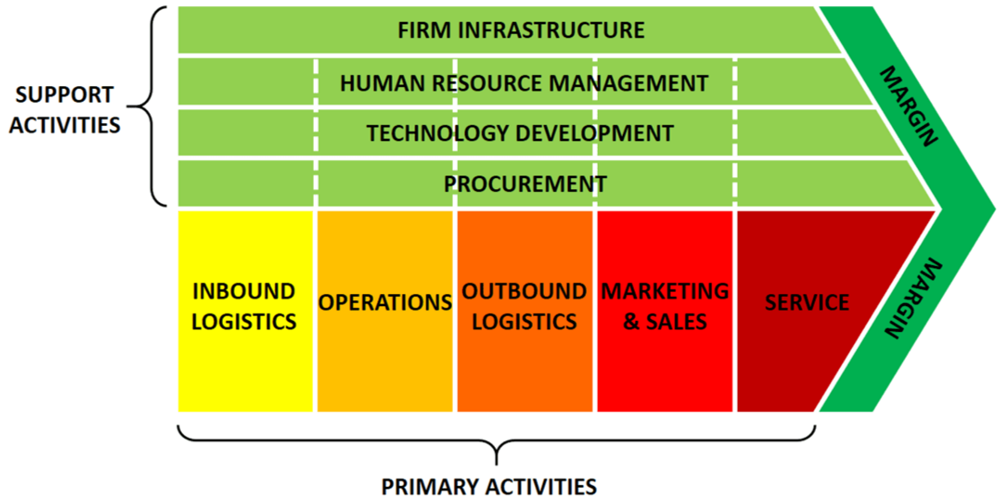
Your CEO says: *'We have a lot of data. I want us to become data-driven.'*
What do you do **on Monday morning**?

Data Strategy Introduction

A data strategy defines how data supports strategic objectives by improving decisions, operations, or differentiation.

Data is not a goal. Data is a strategic asset only if it changes decisions and outcomes.

Data does not replace the value chain. It amplifies specific decisions inside it.



Value Chain Step	Typical Decisions	Data Levers
Operations	How much to produce? When?	Forecasting, optimization
Marketing	Who to target? At what price?	Segmentation, elasticity
Sales	Who to prioritize?	Lead scoring
Supply chain	How much inventory?	Demand prediction
Service	How to reduce churn?	Usage & behavior data

Data Use Case

- A data use case
  - One decision
  - one user
  - one data source
  - one measurable impact

Function	Decision	Data Use Case
Marketing	Who to target	Propensity-to-buy model
Pricing	At what price	Price elasticity model
Ops	How much to produce	Demand forecasting
Finance	Where margin leaks	Cost-to-serve analysis
HR	Who might leave	Attrition prediction

Data Strategy in Marketing – Business Case (1/3)

On online insurer invests heavily in online acquisition: traffic is growing, but revenue stagnates.
Management asks: “Where are we losing value in the acquisition funnel?

1st questions

- What are the main steps between a visit and a customer?
- At which step do you expect the biggest drop?
- Is traffic volume or traffic quality more important?

Data Strategy in Marketing – Business Case (2/3)

Funnel Step	Volume
Website visits	100,000
Leads / signups	9,000
Qualified leads (SQL)	3,600
Paying customers	900

Channel	Leads	SQL
SEO	5,600	2,800
Paid Ads	2,100	500
Social	1,300	300

Cohort	Month 0 (Signup)	Month 1	Month 2
January	100%	62%	48%
February	100%	58%	42%
March	100%	49%	35%

1. Where is the main performance bottleneck in the funnel?
2. Which acquisition channel brings the highest-quality leads?
3. Is growth improving over time or hiding a structural issue?

Data Strategy in Marketing – Business Case (3/3)

1. Felt conversion:

- Visit → Lead = 9%
- Lead → SQL = 40%
- SQL → Customer = 25%

Overall conversion = 900 / 100,000 = 0.9%

- ➢ The biggest value loss happens before qualification
- ➢ The problem is not sales closing, but traffic → lead quality
- ➢ This is primarily a marketing / targeting / landing page issue

2. SQL rate by channel:

- SEO: 2,800 / 5,600 = 50%
- Paid Ads: 500 / 2,100 ≈ 24%
- Social: 300 / 1,300 ≈ 23%

➢ SEO produces **much better leads**

→ Reallocate budget **or** redesign paid acquisition targeting

3. Month-2 retention declines by cohort:

- January: 48%
- February: 42%
- March: 35%

- ➢ **New cohorts retain worse than older ones**
- ➢ Growth is driven by acquisition, not product quality

Data Strategy in marketing - Tools

Tool	Data Collected	Business Questions Answered
Google Analytics	Visits, source, pages	Are we attracting the right traffic?
HubSpot / Salesforce	Leads, status, pipeline	Are leads qualified and converted?
Mixpanel	User actions, activation	Do users understand and use the product?
Tableau / PowerBI	Aggregated KPIs, correlation	Where should management act first?

Data Strategy in Sales – CRM & KPIs

Tool	Business Use
Lead scoring	Rank prospects by probability
Pipeline analytics	Identify stalled stages
Win/loss analysis	Understand objections
Sales cycle tracking	Reduce time-to-close

Data Strategy in Operations

A manufacturing company sells 3 products. Sales teams push all products equally, despite growing volumes, overall margin is declining.

Are all products equally profitable once operations are taken into account?

1. What is the operational margin by product?
2. What operational issues explain this result?

Product	Units Sold	Revenue (€k)	Gross Margin (%)	Gross Profit (€k)
Product A (Standard)	50,000	1,000	45%	450
Product B (Customized)	20,000	800	40%	320
Product C (Complex / Low volume)	10,000	600	35%	210
Total	—	**2,400**	—	**980**

Product	Production Setup (€k)	Logistics (€k)	After-sales & Rework (€k)	Total Ops Cost (€k)
Product A	50	40	30	120
Product B	90	60	70	220
Product C	130	90	160	380

Data Strategy in Operations

Operational Margin

- Product A: 450 – 120 = **+330**
- Product B: 320 – 220 = **+100**
- Product C: 210 – 380 = **–170**

Operational Issues:

- High setup times
- Low utilization
- Customization and rework
- Complex logistics
- High after-sales load

Different tools:

- ERP production & logistics data
- Excel + SQL cost allocation
- Power BI / **Tableau**

Data strategy in HR and IT

Function	Management Question	Data Used	KPIs Calculated	Tools Commonly Used	Decisions Enabled
HR	Are we at risk of losing key employees?	Tenure, performance reviews, salary, absenteeism	Turnover rate, attrition risk score	**Workday, SAP SuccessFactors, Excel**	Retention plans, salary adjustments, succession planning
HR	Which teams have productivity issues?	Headcount, output, overtime	Revenue per employee, absenteeism rate	**Power BI, Excel**	Reallocation, hiring freeze, workload rebalancing
HR	Is hiring efficient?	Time-to-hire, sourcing channel, recruiter load	Time-to-fill, cost per hire	**ATS + Excel**	Process redesign, channel prioritization
IT	Are IT resources used efficiently?	Server usage, licenses, tickets	Utilization rate, cost per user	**ServiceNow, Tableau**	License reduction, infrastructure optimization
IT	Where do incidents slow the business?	Tickets, resolution time, downtime	Incident rate, MTTR (Mean Time to Repair)	**ServiceNow, Power BI**	Process automation, system upgrades
IT	Is cybersecurity risk increasing?	Login logs, incidents, vulnerabilities	Number of incidents, patch delay	**SIEM tools + BI**	Security investment prioritization

Where Data Strategy often fails

Dashboards Without Decisions

- ➢ Dozens of dashboards
- ➢ Beautiful charts
- ➢ No one changes behavior

Rule:
If no decision follows, the KPI should not exist.

KPIs with no owner

- ➢ Everyone sees the KPI
- ➢ No one feels accountable
- ➢ Problems persist

Rule:
- **One owner**
- **One target**
- **One review rhythm**

Over-Engineering Models

- ➢ Complex machine learning models
- ➢ Long development cycles
- ➢ Low adoption by business teams

Rule:
- **If users don't trust or understand the output, the model is useless.**
- **Move to advanced models only when decisions are stable**

How to assess the data maturity of one company?

Dimension	Key Question	Yes / No
Strategy & Decisions	Are the key business decisions clearly identified and supported by data?	
KPIs & Ownership	Does each KPI have one clear owner and a target?	
Data & Tools	Is data accessible, reliable, and actually used by managers?	
Action & Follow-up	Do KPI reviews systematically lead to clear actions?	
Incentives & Culture	Are incentives aligned with data insights and decisions?	